

Amirhossein Rajabpour

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Headline

Bridging ML research and engineering • Machine Learning, Reinforcement Learning, Generalization

Education

- **University of Alberta** 2023 – 2025
M.Sc. in Computing Science, advised by [Levi Lelis](#) & [Sandra Zilles](#) Edmonton, AB, Canada
Thesis: Reevaluating Out-of-Distribution Generalization of Programmatic and Neural Policies through Improved Benchmarking [link](#), GPA: 3.8/4
- **Amirkabir University of Technology (Tehran Polytechnic)** 2018 – 2023
B.Sc. in Computer Engineering (Major: AI, Minor: Computer Networks) Tehran, Iran
GPA (last two years): 3.96/4

Publications

- *Revisiting OOD Generalization in Programmatic RL (ICML 2026)*.
– A. Rajabpour, K. Aghakasiri, S. Zilles, L. Lelis. [\[Preprint-soon\]](#) [\[Code\]](#) [\[Page\]](#)
- *GLASS: Test-Time Acceleration for LLMs via Global-Local Neural Importance Aggregation (Submitted for Neurips 2026)*.
– A. Sattarifard, S. Lavasani, K. Zhang, A. Rajabpour, H. xu, F. Sun, N. Hassanpour, C. Gao. [\[Preprint\]](#)
- *Common Benchmarks Undervalue the Generalization Power of Programmatic Policies (RLC 2025 Programmatic RL Workshop)*.
– A. Rajabpour, K. Aghakasiri, S. Zilles, L. Lelis. [\[Preprint\]](#)

Technical Skills

- **Programming:** Python, C/C++, Java, Kotlin, Octave
- **ML/AI:** PyTorch, TensorFlow, Keras, scikit-learn, fastai, OpenCV
- **Data Science:** NumPy, Pandas, Matplotlib, Seaborn
- **Data/DB:** MySQL, PostgreSQL, MongoDB
- **Web:** Django, Flask, HTML, CSS
- **Tools/OS:** Git, Docker, Jira, Selenium, XAMPP, Linux, Windows, macOS

Work Experience

- **AI Researcher**, Huawei Technologies Canada Oct 2025 – Present
– Optimize and deploy large language models on mobile NPUs and CPUs, focusing on compilation, sparsification, and hardware-aware performance tuning.
– Deployed and configured Slurm-based GPU clusters, including partition setup, fair-share scheduling, and resource allocation policies for large-scale ML experiments.
– Demoed OpenClaw as an agent-style system that can plan, use tools, and carry out multi-step tasks on its own.
Tech: Python, C++, PyTorch, ONNX, Bash, Compiler, Google LiteRT, vLLM
- **Research Assistant**, University of Alberta May 2024 – Jul 2025
– Refined neural policies (RL agents) through targeted architectural and training adjustments, enabling them to match or outperform programmatic policies in generalization on key benchmarks.
– Proposed new benchmarks highlighting strengths of programmatic policies (e.g., memory-based reasoning).
– Built end-to-end training/evaluation pipelines.
– Used **Large Language Models** to iteratively generate and refine interpretable policies in settings where RL underperforms.
– Work accepted at [RLC 2025 Programmatic RL workshop](#).
Tech: Python, PyTorch, TensorFlow, Bash
- **Technical Advisor**, Alberta Machine Intelligence Institute (Amii) Sep 2024 – Oct 2024
– 1-month project on advanced AI/ML methods to enhance real-time intrusion detection and anomaly detection in high-speed network storage infrastructures.
– Focused on threat-detection accuracy and scalability in multi-cloud settings.
- **Machine Learning Engineer**, R&D Dept., Crouse PJS Co. Oct 2021 – Jan 2022
– Designed lightweight industrial ML for detecting malfunctioning LEDs: localization (OpenCV), fuzzy C-means clustering of light pixels, luminance/wavelength extraction, and per-color regression modeling.
Tech: Python, OpenCV, Scikit-learn
- **DevOps Engineer Intern**, Graph Co. Nov 2020 – Apr 2021

- Hands-on experience with Docker and cloud-native tooling.

Additional Research Projects

- **Individual Study Course Project**, University of Alberta Jan 2023 – Apr 2024
 - Implemented hill climbing (with exploration) and a genetic algorithm to improve “[Unveiling Options with Neural Decomposition](#)”. Supervisor: Levi Lelis.
- **Reinforcement Learning 1 Course Project**, University of Alberta Sep 2023 – Dec 2023
 - Compared generalization of expert (concentrates resources on a single task) vs. generalist (splits resources across tasks) agents (PPO, DQN, APPO) across MiniGrid and budget regimes. [\[Project Report & Code\]](#)
- **Bachelor’s Thesis: Face Aging Platform**, Amirkabir University Mar 2023 – May 2023
 - Developed a face-aging platform using generative models (e.g., CycleGAN, reversible models). Supervisor: Mohammad Rahmati. [\[Code\]](#)
 - Integrated multiple pretrained models with distinct dependencies into a unified system using Docker Compose, deploying each in separate containers and orchestrating communication across them.
- **Research Assistant: Portfolio Allocation with RL & GNNs**, Amirkabir University Feb 2022 – Jun 2022
 - Implemented a graph convolutional network over time-series for asset allocation. Supervisor: Hamed Farbeh.

Teaching Assistant

- **University of Alberta**
 - CMPUT 365: Introduction to Reinforcement Learning | Instructor: Marlos Machado Fall 2025
 - CMPUT 366: Search & Planning in AI | Instructor: Levi Lelis Fall 2024, Winter 2025
 - CMPUT 291: Intro to File & Database Management | Instructors: Davood Rafiei, Arash Nobari Fall 2023, Winter 2024
- **Amirkabir University of Technology**
 - Principles of AI | Instructor: Mahdi Javanmardi Fall 2022
 - Cloud Computing | Instructor: S. Ahmad Javadi Fall 2022
 - Internet of Things | Instructor: Siavash Khorsandi Fall 2022
 - Algorithm Design | Instructors: A. Bagheri, S. Shirali-Shahreza Winter 2022, Fall 2021, Winter 2021

Honors and Awards

- Master’s admissions: Fully funded offers from UAlberta (CS, ECE, Radiology & Diagnostic Imaging) & Western Ontario (CS) 2023
- Bachelor’s Scholarship: 4-year scholarship, Amirkabir University of Technology 2019–2023
- Nationwide Mathematics University Entrance Exam (Iran): Top 1% among 140,000+ applicants 2018

Graduate Courses

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| • Reinforcement Learning 1 (Marlos Machado) | • Modelling Strategic Behavior (James Wright) |
| • Reinforcement Learning 2 (Rich Sutton) | • Machine Learning (Lili Mou) |

Language Skills

- Persian: Native
- English: TOEFL iBT 109 (R 26, L 28, S 26, W 29)
- German: Professional Working Proficiency (B2)
- French: Beginner (A1)